

TaskWeaver Modeling Language

First conceptualization of our language in the 'Metamodeling' course

University of Fribourg, 12.11.2025

Objectives & Purpose

1

Enable visual design of automation & data-processing workflows

2

Bridge models to execution via exportable artifacts (e.g., JSON)

3

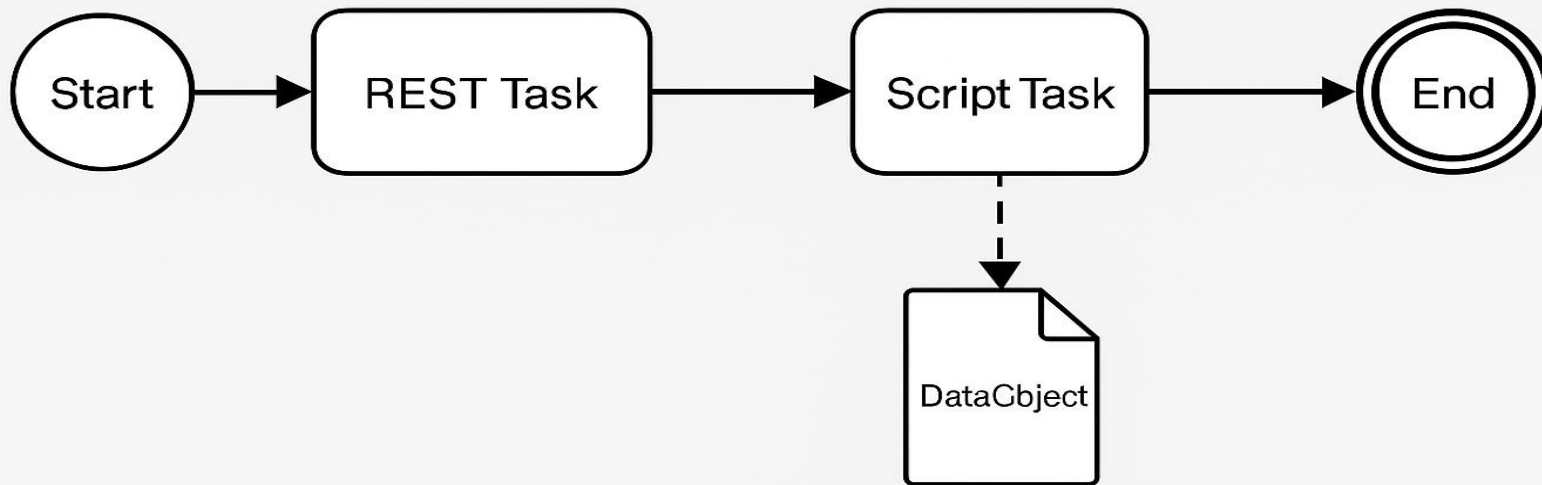
Provide lightweight validation & analysis for correctness and timing

Conceptualization — Domain Overview

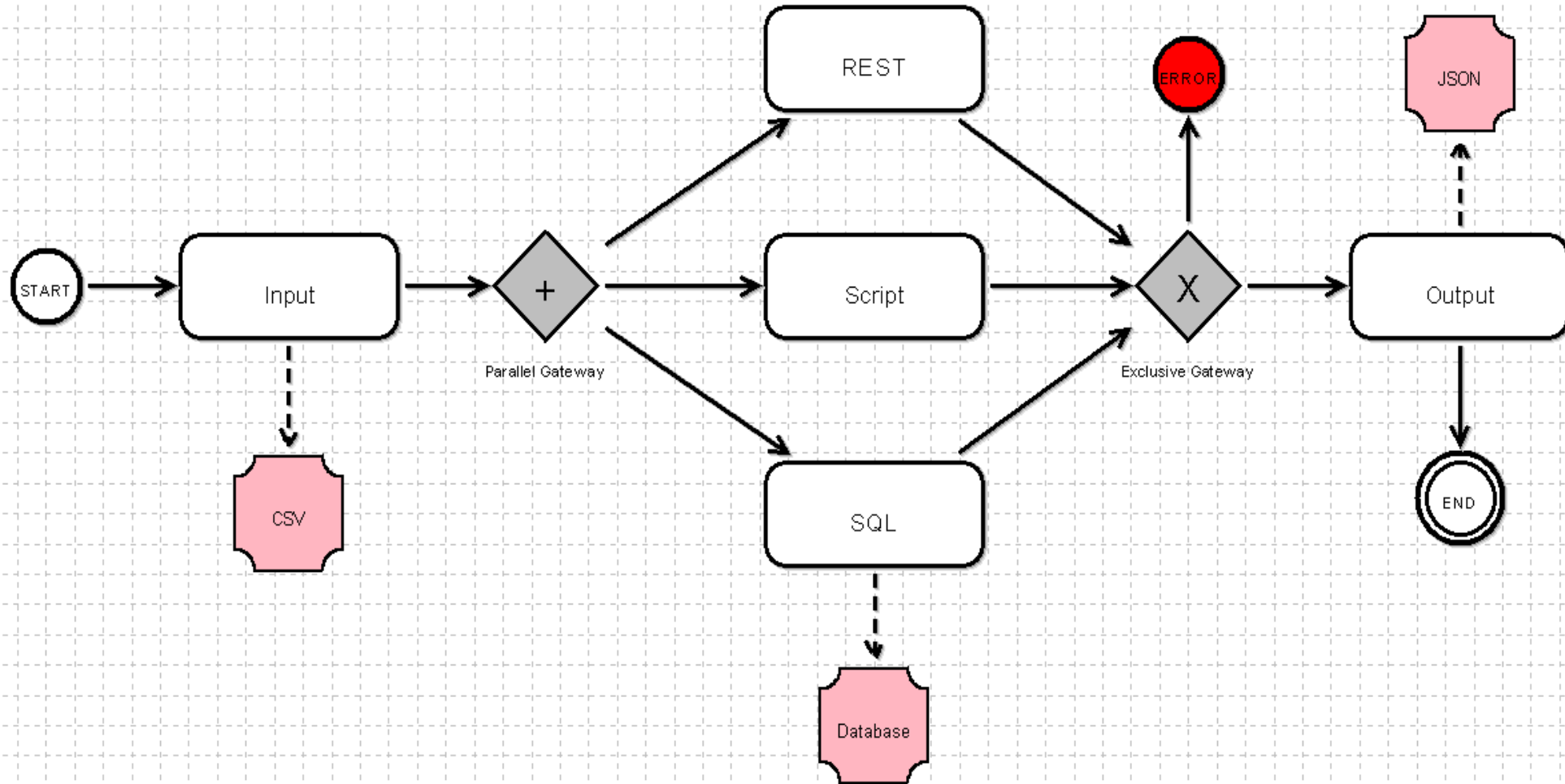
- **Core elements:**
Task, DataObject, Event (Start / End / Error), *Gateway* (Exclusive / Parallel)
- **Relations:**
SequenceFlow (control), *DataFlow* (data)
- **Typical tasks:**
REST, Script, SQL, or ML; data as files, tables, datasets



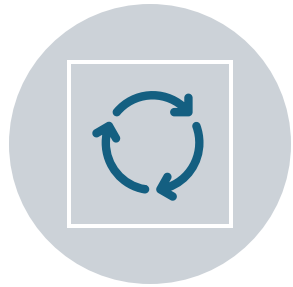
First metamodel example (Abstract Syntax)



Improved metamodel



Next Steps



Add validator: Start/End checks, reachability, simple cycle error detection



Implement JSON export (model-to-text via ADOScript)



Refine semantics by creating and testing more models



Request feedback sessions and improve the language further



Thank you.

Questions? 🤪